

flex-direction

A B C

C B A

justify-content

start

center

flex-end

space-between

space-evenly

align-items

stretch

center

flex-wrap: wrap

wrap

1 2 3 4 5

flex item sizing

flex:1 flex:2 flex:1

margin-top: auto (push to end)

Header

Content

Footer (margin-top:auto)

CONTAINER PROPERTIES

display: flex

flex-direction

row → left to right (default)

row-reverse ← right to left

column ↓ top to bottom

column-reverse ↑ bottom to top

justify-content

(main axis alignment)

flex-start

flex-end

center

space-between

space-around

space-evenly

align-items

(cross axis alignment)

stretch Fill container height (default)

flex-start Align to start

flex-end Align to end

center Center on cross axis

baseline Align text baselines

flex-wrap

nowrap Single line — can overflow (default)

wrap Wrap to next line

wrap-reverse Wrap in reverse order

gap: 16px | row-gap col-gap

flex-flow: row wrap (direction + wrap)

align-content: space-between (multi-line)

ITEM PROPERTIES

flex shorthand

flex: grow shrink basis

flex: 1 = 1 1 0% — grow equally from zero

flex: auto = 1 1 auto — grow from content size

flex: none = 0 0 auto — rigid, never changes

flex: 0 0 200px fixed width — never grows or shrinks

flex-grow: 0 | 1 0=stay 1=grow into free space

flex-shrink: 0 | 1 0=rigid 1=shrink if overflow

flex-basis: auto | 0 | 200px starting size before growth

align-self: center | flex-end override container align-items

order: 0 | -1 | 1 -1=before 1=after 0=default

KEY PATTERNS

Perfect center:

display: flex; place-items: center

Sticky footer:

body: flex-col; main: flex: 1

Navbar:

justify: space-between; align: center

Card footer down:

flex-col; footer: margin-top: auto

Equal width tabs:

children: flex: 1

Fixed sidebar:

sidebar: flex: 0 0 280px; main: flex: 1

Reverse order:

flex-direction: row-reverse

justify-content

= aligns along MAIN axis

In row → horizontal

flex: 1 vs flex: auto

= aligns along CROSS axis

flex: 1 = flex: 1 1 0%

basis=0: siblings share space EQUALLY

flex: auto = flex: 1 1 auto

basis=content: bigger item gets MORE

CSS GRID

Basic Grid: 3 columns x 2 rows

1 2 3

4 5 6

fr unit: 1fr 2fr 1fr

1fr 2fr 1fr

grid-template-areas (Named Areas)

header

sidebar main aside

footer

grid-column: span / grid-row: span

span 2 cols 2 rows

B C

D E

auto-fill: minmax(280px, 1fr)

No media queries — auto responsive!

Wide (4 cols)

Mid (2 cols)

Mobile (1 col)

place-items: center

Centered items in each cell

1 2 3

CONTAINER PROPERTIES

display: grid

grid-template-columns

200px 1fr Fixed + flexible

repeat(3, 1fr) 3 equal columns

1fr 2fr 1fr 25% / 50% / 25%

repeat(auto-fill, Fill row with cols minmax(280px, 1fr)) min 280px each

repeat(auto-fit, Collapse empty cols minmax(200px, 1fr)) items stretch to fill

grid-template-rows

64px 1fr 80px Fixed header/footer

auto Size to content

repeat(3, 100px) 3 fixed rows

minmax(100px, auto) Min 100px, grows

grid-template-areas

'header header'

'sidebar main'

'footer footer'

gap: 24px | 16px 24px row-gap col-gap

row-gap: 16px rows only

column-gap: 24px columns only

justify-items: start | end | center | stretch items on row axis

align-items: start | end | center | stretch items on col axis

place-items: center justify + align together

justify-content: start | end | center tracks on row axis

align-content: start | end | center tracks on col axis

place-content: center tracks both axes

grid-auto-flow: row | column | dense implicit item placement

grid-auto-rows: 100px implicit row height

grid-auto-columns: 200px implicit col width

ITEM PROPERTIES

grid-column

1 / 3 From line 1 to line 3

1 / -1 From first to LAST line

span 2 Span 2 columns

span 3 Span 3 columns

grid-row

1 / 3 From row line 1 to 3

span 2 Span 2 rows

grid-area: sidebar Use a named template area

grid-area: 1/1/3/3 row-start/col-start/row-end/col-end

justify-self: start | end | center | stretch item on row axis

align-self: start | end | center | stretch item on col axis

place-self: center justify-self + align-self

GRID LINE NUMBERS

1 2 3 4

1 2 3

1/2 to 2/3

NAMED AREAS — FULL CODE

.layout {

display: grid;

grid-template-columns: 200px 1fr 200px;

grid-template-rows: 64px 1fr 64px;

grid-template-areas:

'header header header'

'sidebar main aside'

'footer footer footer';

min-height: 100vh;

gap: 0;

}

header { grid-area: header; }

.sidebar { grid-area: sidebar; }

main { grid-area: main; }

aside { grid-area: aside; }

footer { grid-area: footer; }

RESPONSIVE GRID CODE

/* NO MEDIA QUERIES NEEDED */

.cards {

display: grid;

grid-template-columns:

repeat(

auto-fill,

minmax(280px, 1fr)

);

gap: 1.5rem;

}

/* auto-fill: max columns that fit */

/* minmax: min 280px, max 1fr */

/* Result: 4col→3col→2col→1col */

/* all automatically, no @media! */

/* auto-fill vs auto-fit */

/* auto-fill: keeps empty tracks */

/* auto-fit: collapses empty cols */

KEY PATTERNS

Dashboard layout:

grid-areas with named sections

Card grid:

auto-fill + minmax — no @media

Full-width item:

grid-column: 1 / -1

Center in cell:

place-items: center on container

Center one item:

place-self: center on item

Equal columns:

repeat(N, 1fr)

Fixed + fluid:

240px 1fr (sidebar + main)

Overlap items:

same grid-area or overlapping lines

Subgrid:

grid-template-rows: subgrid

GRID vs FLEXBOX

Flexbox

Grid

Dimensions

1D only

2D rows+cols

Best for

Navbars, rows

Layouts, dashboards

Items driven by

Content

Structure

Overlapping

Hard

Easy

Responsive

flex-wrap

auto-fill/fit

fr UNIT EXPLAINED

fr = fraction of FREE space after fixed cols

1fr 2fr 1fr = 4 parts total:

first col = 1/4 = 25%

middle col = 2/4 = 50%

Unlike %: fr accounts for gap space

CSS FLEXBOX Container Properties Item Properties flex: 1 vs flex: auto Patterns Axis Rule

CSS GRID Container Properties Item Properties Named Areas Visual Examples Grid vs Flex fr Unit

Akshay Chavhan